

Enhancing mobility management in 5G networks using deep residual LSTM model

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HIGHLIGHTS

- Efficient use of resources and seamless handovers achieved through advanced algorithm.
- Significantly reduces network congestion and latency for enhanced user experience.
- Employs machine learning techniques for intelligent decision making.
- Optimizing mobility management for the next generation of wireless networks.
- Achieving better network performance and efficiency with advanced techniques.

ARTICLE INFO

Keywords:

5G networks
Mobility management
Handover optimization
LSTM
Resource Allocation

ABSTRACT

Mobility management is an essential component of 5G networks to provide mobile users with seamless connectivity and efficient cell transition. However, increasing user mobility, device density, and the diversity of service requirements all pose significant challenges to achieving optimal mobility management. This article describes a novel method for improving mobility management in 5G networks that employs a deep residual Long Short-Term Memory model. Deep learning and LSTM, a type of recurrent neural network, are used in the proposed model to identify temporal dependencies and patterns in user mobility data. The model learns to predict future user locations and mobility patterns by training on historical mobility data, allowing for proactive resource allocation and handover decisions. We incorporate residual connections into the LSTM architecture, inspired by the residual learning framework, to address the inability of traditional LSTM models to capture complex temporal dynamics. This allows the model to effectively incorporate long-term dependencies and improves prediction accuracy. Furthermore, we incorporate the mLSTM model into the mobility management framework of 5G networks. The model continuously obtains real-time user location updates and predicts future user positions, allowing for proactive handover decisions. The network can optimize resource allocation, reduce handover latency, and improve user experience by leveraging anticipated mobility patterns. We test the proposed method by simulating it extensively with real-world mobility traces. The results show that the mLSTM model accurately predicts user mobility and outperforms conventional methods in transition performance. The model is not affected by changing network conditions, user mobility patterns, or service specifications.

1. Introduction

Significant advancements in wireless communication technology have resulted from the introduction of 5G networks, enabling faster data rates, lower latency, and increased network capacity. Mobility

management, which involves efficiently managing the movement of mobile users as they transition between different cells or access points, is a crucial aspect of 5G network electronics [1]. Mobility patterns involving user movements can be highly dynamic and unpredictable, making it difficult to accurately predict their next location. This can lead

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<https://doi.org/10.1016/j.asoc.2024.112103>

Received 23 October 2023; Received in revised form 12 June 2024; Accepted 3 August 2024

Available online 14 August 2024

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to inefficient resource allocation and handover decisions. In densely populated urban areas and at large-scale events, the mobility density can be extremely high. This causes an increase in handoffs and resource allocation demands, straining the network infrastructure. To avoid congestion and maintain seamless connectivity, it is essential to effectively manage handovers in such dense environments [2]. Depending on their individual preferences, activities, and preferences, various users may have distinct mobility patterns. Some users may have regular commuting patterns, whereas others may have random and irregular movements [3]. It is a significant challenge to accommodate these diverse mobility patterns and customize mobility management solutions for various user groups [4]. However, the increasing mobility of users, the proliferation of connected devices, and the varying requirements of different services present significant obstacles to achieving optimal mobility management in 5G networks [5]. Traditional approaches like Adaptive Time-to-trigger (TTT), Inter-slice mobility management, reinforcement learning algorithms and Sailfish-based Distributed IP Mobility Management (SbDMM), and few other algorithms operates based on handover mechanisms and resource allocation algorithms may have difficulty adapting to the dynamic nature of user mobility patterns and fluctuating network conditions [6]. Therefore, 5G networks have an urgent need for innovative solutions that can improve mobility management [7]. The application of novel strategies and cutting-edge technologies, such as machine learning, deep learning, and data analytics, is necessary for addressing these obstacles and issues associated with user mobility patterns [8]. Using these techniques, network operators can develop efficient mobility management strategies that adapt to the dynamic nature of user mobility, optimize resource allocation, reduce handover disruptions, and improve the overall user experience in 5G networks [9,10]. Deep learning techniques have emerged as effective methods for analyzing and modeling intricate data patterns [11]. Recurrent Neural Networks (RNNs), such as Long Short-Term Memory (LSTM), have been utilized successfully in a variety of domains to capture temporal dependencies in sequential data [12]. These networks can make predictions based on historical data and learned patterns. Utilizing deep learning techniques can enable accurate predictions of future user locations and mobility patterns in the context of mobility management [13]. By anticipating user movements, proactive decisions can be made regarding resource allocation and handoffs, resulting in enhanced network efficiency and user satisfaction. However, conventional LSTM models may have difficulty capturing the complex temporal dynamics present in user mobility data [14]. Complex mobility patterns and long-term dependencies can present difficulties for standard LSTM architectures, thereby limiting their predictive capabilities [15]. The concept of residual learning, inspired by the success of residual networks in computer vision tasks, can be applied to circumvent this limitation. Residual connections in LSTM architectures allow the model to effectively incorporate long-term dependencies and capture complex temporal dynamics, thereby increasing the precision of predictions and enhancing the performance of mobility management [16]. The proposed research seeks to bridge the gap between deep learning techniques and 5G network mobility management. Using a deep residual LSTM model, the study aims to develop a novel method for accurately predicting the future locations and mobility patterns of users [17]. The model will be trained using historical mobility data, allowing it to identify temporal dependencies inherent in user movements. By integrating the model predictions into the existing mobility management framework of 5G networks, proactive resource allocation and handover decisions can be made, resulting in enhanced network performance and user experience [18]. Enhancing mobility management in 5G networks has substantial benefits. Proactive resource allocation can guarantee the efficient utilization of network resources, resulting in increased capacity, decreased congestion, and improved Quality of Service (QoS) for users [19]. Additionally, proactive handover decisions based on accurate predictions can minimize handover latency, enabling seamless connectivity between cells during user transitions. By incorporating the deep residual

LSTM model into the framework for mobility management, network operators can adapt to the dynamic nature of user mobility, resulting in improved network performance and increased user satisfaction [20]. The article introduces a novel approach by combining deep learning with a residual LSTM model for mobility management in 5G networks [21]. This model is designed specifically to capture temporal dependencies and patterns in user mobility data, enabling accurate predictions of future user locations and mobility patterns.

The proposed research is novel in that it identifies performance issues and intelligently allocates resources to address them, resulting in an automated and efficient 5G network management solution.

- The LSTM model, with its deep residual architecture, has been introduced as a powerful tool for mobility management in 5G networks. This model enables more accurate prediction of user mobility behaviour, making it easier to manage how users move across networks and subnetworks.
- The deep residual architecture also enables better model generalisation for various multidimensional time-series mobility datasets. This enables the model to be deployed in a variety of environments with varying network conditions and user mobility patterns, increasing its credibility and robustness.
- The proposed model makes use of the capacity of both LSTM and recurrent neural networks, allowing the model to learn the short and long-term correlations of user mobility and thus increase the model's accuracy.

This model is specifically designed to capture temporal dependencies and patterns in user mobility data, allowing accurate predictions of future user locations and mobility patterns.

- The contribution of this study is a novel method for improving mobility management in 5G networks using mLSTM models. The method improves resource allocation, reduces handover latency, and improves user experience in dynamic and diverse network environments by utilising advanced deep learning techniques.
- The proposed method enables proactive resource allocation and handover decisions by utilising mLSTM model predictions. This enables the network to anticipate user movements and optimise resource allocation ahead of time, reducing handover latency and improving overall user experience.
- The mLSTM model is integrated into the existing mobility management framework in 5G networks. It constantly receives real-time user location updates and makes predictions, allowing the network to make informed decisions based on anticipated mobility patterns.

The remaining parts of the research paper have constructed the following. Chapter 2 discusses the details of related works. Chapter 3 provides the network design, proposed model, and algorithm. Chapter 4 shows the results and discussion between the existing and proposed models. Chapter 4 shows the ablation study. Finally, chapter 5 provides the conclusion and future scope of the proposed research.

2. Literature review

Karmakar, R., et al. [22] has discussed a novel intelligent handover procedure that adds adaptive time-to-trigger and hysteresis margin to the traditional handover algorithm to improve signal robustness and user experience. This handover procedure considers more criteria than the current signal quality before handing over. Also, it takes into account parameters such as the target cell's proximity, network load, user experience, or application-specific preferences, thus providing an improved experience for moving users. In addition, the adaptive time-to-trigger and hysteresis margin allows for greater accuracy in the handover computations on both the user and network side, thereby improving system efficiency and reducing the handover failure rate.

Sajjad, M. M., et al. [23] has discussed a service that enables mobile devices to remain connected to their applications and services when switching between different slices or data plans. ISMM supports the migration between slices during a mobility event, providing an uninterrupted connection to the user's remote applications and services. Tanveer, J., et al. [24] has discussed an essential artificial intelligence technique that can solve complex optimization problems in dynamic complex environments. It has been widely used for solving various problems in various fields, such as communications, finance, and healthcare. Due to its excellent performance, reinforcement learning algorithms have become attractive for handover management in 5G ultra-dense small cell networks. Sharma, V. K., et al. [25] has discussed a technique that uses machine learning algorithms to analyze data collected from 5 G-enabled intelligent devices in the home in order to identify potential threats or events of interest. The algorithm(s) would use pattern recognition, anomalies, and trends to detect these threats. The use of 5G technology would allow for higher bandwidths, faster speeds, and more reliability than earlier technologies used for home security systems. By quickly and accurately identifying threats and events of interest, this technique could enable more intelligent and more secure home security systems. Chen, Y., et al. [26] has discussed a decision-making strategy that considers a network's dynamics. It is a repeated game between a leader and follower, with the leader setting the prices for resource allocation and the follower trying to maximize their gains. The leader's actions influence the follower's decisions, and in the end, the leader's goal is to achieve an equilibrium that benefits both parties. In mobile edge computing, the leader can be considered the service provider setting the resource allocation and pricing policies. In contrast, the follower can be considered the user trying to maximize their gain by carefully choosing the appropriate resources for their mobile application. This game-theoretic approach can be used to optimize the resource allocation and pricing strategies for different types of users and workloads, ultimately resulting in more efficient use of shared resources. Alhashimi, H. F., et al. [27] has discussed a set of techniques for efficiently managing resources within a network. These techniques allow network operators to optimize the usage of their network resources to meet their customers' changing needs and maximize their profit potential. This includes managing and controlling the use of bandwidth, radio access links, and other resources to allocate resources among applications for the best performance possible. It also means optimizing network performance regarding latency, reliability, and security. It also involves providing efficient signaling and control mechanisms to manage the congestion of the network efficiently. Additionally, 6 G heterogeneous networks require resource management techniques to support different vendors and technologies and effectively integrate them into the network.

Liu, L., et al. [28] has discussed an advanced algorithm used to manage network resources efficiently. This algorithm considers traffic patterns, latency, bandwidth, network capacity, user priorities, etc. It uses various algorithms like cloud bursting, traffic engineering, and load balancing to determine resource utilization properly and offer optimized services to the users. The algorithm also looks for effective ways to utilize cloud-based resources to ensure optimized service delivery. Raja, S. K. S., et al. [29] has discussed the CAC schemes are used in wireless cellular networks to manage traffic and ensure that the limited network capacity is utilized efficiently. CAC is used to regulate the number of users that are allowed to establish a connection to the network at any given time. CAC schemes determine how many new calls are accepted for processing based on network load and other factors to ensure that the network continues functioning correctly and that all established connections remain intact. CAC ensures that a limited amount of resources are devoted to each user and that only one user can consume up to their fair share of resources. Zhang, T., et al. [30] has discussed a form of distributed computing that allows data to be processed and stored by a network of connected devices rather than a centralized data center. Edge computing is attractive to industrial and commercial entities because of

its ability to move processes and data closer to end users or devices that generate the data. The industrial Internet of Things (IIoT) can realize several benefits, including cost savings, improved privacy and security, and faster data retrieval speeds by processing data near its source. Edge computing can help facilitate more intelligent operation and analytics, allowing for more proactive decision-making and improved operational efficiency. It can help reduce downtime, optimize maintenance schedules, and even automate specific processes. For example, edge computing can monitor factory systems for potential problems, keep track of machinery performance, and coordinate inventory. So, Edge computing is critical in industrial internet applications, as it can help to improve accuracy, reduce latency, and provide a more secure and reliable communication system. Vinya, V. L., et al. [31] has discussed an Analysis of Cognitive Radio Ad-hoc Networks (CRANs) studies the various technologies that enable intelligent, self-organizing, and self-learning radio networks. CRANs are systems composed of several nodes that allow users to access the same radio spectrum on a "cognitive" or "semi-autonomous" basis. Cognitive radio networks are a promising solution for addressing the problems associated with spectrum scarcity since they can increase network efficiency and improve radio spectrum utilization. Cognitive radio ad-hoc networks aim to provide an efficient solution for dynamic spectrum access. Al-Dulaimi, O. M. K., et al. [32] has discussed the Performing Strategic Spectrum Sensing Study for Cognitive Radio Networks is studying radio spectrum utilization to understand better how spectrum resources are being used and what strategies can be implemented to improve the users' ability to access dynamic resources within the spectrum. It involves collecting data on spectrum use, analyzing the data, and then planning a strategy that effectively utilizes the available space. This strategy could include frequency-dependent trade-offs, spectrum-sharing agreements, and the creation of dynamic spectrum access protocols. The primary goal of this process is to ensure that spectrum access is as efficient and equitable as possible. Prabhas, M. U., et al. [33] has discussed a process whereby CRSNs can cooperate and communicate with each other to enhance the network's lifetime. By utilizing CSS, CRSNs can share their sensed spectrum information and coordinate their spectrum usage to achieve maximum efficiency in energy consumption, traffic distribution, spectrum utilization, and interference mitigation. CRSNs can use this combined information to identify which channels are more suited for their communication and will have minor interference from the environment or other sensor networks. Yuan, S., et al. [34] has discussed the Software-defined intelligent satellite-terrestrial integrated networks are networks that combine space-based assets from both satellite and terrestrial operators. They are designed to improve user service availability, increase reliable and secure data transmission, and enable faster and more efficient communication. They are typically used for broadband-related applications like data centers, business networks, and telemedicine services. These networks can improve performance, scalability, and security by leveraging the best of both worlds.

Logeshwaran, J., et al. [35] has discussed an the utilization of radio access network resources while minimizing the energy consumption of the devices involved. Du, K., et al. [36] has discussed a way of designing a 5G core network where a user's session state is transmitted over multiple messages that do not have to be tracked and stored. This design reduces latency, increases scalability, and removes single points of failure, making it ideal for cloud-native 5G networks. The design relies on microservices and containerization to allow for multiple distributed stateless services that can handle the different 5G session states without keeping track of them. The design can also enable real-time analytics and dynamic network slicing. Hu, J., et al. [37] has discussed a high-quality satellite communication link to an area with little or no terrestrial coverage, such as remote locations or areas with rugged terrain. This method uses two or more satellites in the same orbit plane, each covering a different region so that the entire area is in a constant range. The satellites are configured so that each satellite is constantly in range of the other, ensuring no gaps in coverage. The irregular shapes of

the coverage area prevent gaps between the satellite beams, providing a seamless connection over the entire region. Ramesh, G., et al. [38] has discussed an advanced research and development algorithm to ensure reliable 5G communication network administration. It combines machine learning algorithms, automation techniques, strategic network analysis, and other tools to optimize 5G network performance, increase service reliability, and minimize network latency. SNMA-5 G identifies performance issues and intelligently allocates resources to address them, providing an automated and efficient 5G network management solution. Table 1 shows the comprehensive analysis of related works.

3. Proposed model

To improve mobility management, the system model for a 5G environment with user mobility management integrates the proposed mLSTM model into the existing 5G network infrastructure. The model (Fig. 1) predicts user locations and mobility in the future, allowing for proactive resource allocation and handover decisions.

- Base Stations (BS): These are the 5G network access points that provide coverage and connectivity to mobile users.
- User Equipment (UE): Mobile devices used by users to connect to a 5G network.
- Core Network (CN): The core network manages base station mobility, authentication, and communication with external networks.
- Mobility Management Entity (MME): This entity is responsible for tracking mobile users' locations and managing mobility-related procedures.
- Handover Decision: When a user predicted location indicates that a handover is about to occur, the mobility management module initiates a handover decision and selects the best target base station for the handover.
- Resource Allocation: To ensure uninterrupted service during handovers, the module allocates appropriate resources to the user based on predicted mobility patterns and current network conditions.
- User Location Updates: The mobility management module receives continuous real-time user location updates from the network.

The mLSTM model can enhance collaboration by receiving real-time user location updates and predicting future mobility. Integrated into the 5G infrastructure, it facilitates proactive handover decisions and resource allocation. The model interacts with Base Stations, Core Network, and Mobility Management Entity, enabling seamless coordination for optimized mobility management in the dynamic 5G environment.

3.1. Network model

Mobility management techniques are incorporated into the network model for a 5G environment with mobility user management to ensure seamless connectivity and an efficient transition between cells. The network model and equations for managing mobile users are as follows:

- User Location: $L_t = (x_t, y_t, z_t)$ represents the user location at time t , where x_t , y_t , and z_t are the user x , y , and z coordinates, respectively.
- User Velocity: $V_t = (v_x^t, v_y^t, v_z^t)$ represents the rate of change of the user location over time at t .
- Handover Threshold: When the user distance from the serving BS exceeds a predefined threshold, represented by dt , a handover decision is made.
- Distance Calculation: The distance (d) between the user location and a specific BS can be calculated using the Euclidean distance formula.

$$d = \sqrt{(x_t - x_{BS})^2 + (y_t - y_{BS})^2 + (z_t - z_{BS})^2} \quad (1)$$

Table 1
Comprehensive analysis.

Author	Year	Advantage	Limitation
Alhashimi, H. F., et al. [27]	2023	Efficient use of network resources, optimizing performance and reducing costs in a complex environment with diverse devices and connectivity options.	Inadequate coverage of real-world deployment scenarios and practical implementation challenges.
Prabhas, M. U., et al. [33]	2023	More accurate spectrum sensing and improved network coverage for longer periods of time.	Dependence on cooperative spectrum sensing and lack of consideration for individual sensors' capabilities and spectrum availability.
Logeshwaran, J., et al. [35]	2023	The model improves energy efficiency and optimizes resource allocation for D2D communication in 5 G wireless personal area networks.	The model may not consider all possible network scenarios and user behaviors, leading to suboptimal resource allocation in certain cases.
Ramesh, G., et al. [38]	2023	The algorithm increases efficiency and mitigates human errors, allowing for faster and more accurate management of 5 G networks.	Potential lack of adaptability or effectiveness in handling complex and unpredictable network scenarios.
Karmakar, R., et al. [22]	2022	The proposed model incorporates adaptive time-to-trigger and hysteresis margin, improving overall efficiency and user experience.	Complexity of implementation and potential impact on network resources may hinder practical adoption.
Sajjad, M. M., et al. [23]	2022	It allows for seamless handover between network slices, providing improved user experience and efficient utilization of resources.	The lack of standardization and interoperability among different network slices may hinder seamless mobility across slices.
Tanveer, J., et al. [24]	2022	Efficient and adaptive handover decisions can be made in real time without relying on predefined rules or complicated handover policies.	Susceptibility to environmental changes or unpredictability of user behavior leading to suboptimal handover decisions.
Sharma, V. K., et al. [25]	2022	Efficient use of 5 G technologies improves home security by optimizing algorithms for faster response and better security coverage.	It may be a lack of flexibility in adapting to new or changing security threats in the smart home environment.
Vinya, V. L., et al. [31]	2022	It helps optimize spectrum usage and enable efficient communication by dynamically adapting to changing network conditions and allocating spectrum resources accordingly.	Not accounting for practical implementation challenges and performance metrics in real-world scenarios.
Al-Dulaimi, O. M. K., et al. [32]	2022	Improved efficiency and utilization of available spectrum resources for better connectivity and increased network performance.	Limited scope which may not cover all potential scenarios and may not provide comprehensive solutions for practical implementation.
Yuan, S., et al. [34]	2022	Flexibility and seamless integration between satellite and terrestrial networks for enhanced connectivity and network performance.	Limited functionality due to dependence on software and potential for technical failures or vulnerabilities.

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Table 1 (continued)

Author	Year	Advantage	Limitation
Du, K., et al. [36]	2022	Increased scalability, flexibility, and reliability due to independent processing and storage of messages across the distributed network.	Increased network latency due to the frequent request and response messages exchanged between client and server.
Hu, J., et al. [37]	2022	Its ability to provide seamless and uninterrupted coverage in a flexible manner for high throughput satellites.	Simplistic model may be insufficient to capture complex real-world factors and may not apply to all scenarios.
Liu, L., et al. [28]	2021	Efficient allocation of communication resources to improve speed and connectivity for high speed rail passengers.	It could include scalability issues, reliance on internet connection, and lack of compatibility with outdated technology.
Raja, S. K. S., et al. [29]	2021	Efficient spectrum utilization and better quality of service for mobile users.	Limited scalability due to lack of dynamic prediction and management of varying network conditions.
Zhang, T., et al. [30]	2021	Faster data processing and reduced latency, resulting in improved efficiency and productivity for industrial operations.	Additional infrastructure and maintenance costs may limit the widespread adoption and scalability of edge computing in industrial settings.
Chen, Y., et al. [26]	2020	Better resource allocation and pricing in edge computing through a leader-follower decision-making framework.	Limited applicability to real-world scenarios due to its assumption of perfect knowledge and control of users and resources.

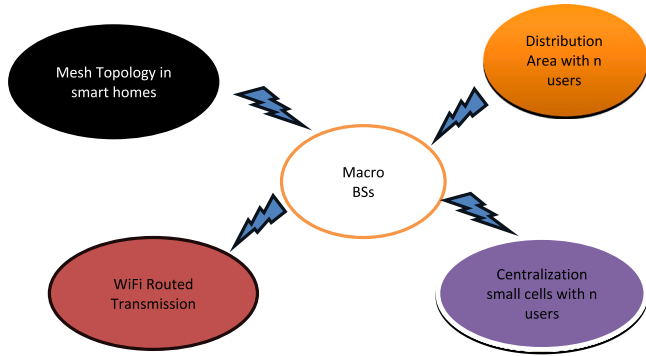


Fig. 1. 5 G system architecture.

- **Handover Decision:** If $d > d_t$, a handover decision is made to switch the user to a target BS.
- **User Demand:** D_t represents each user's unique demand requirements, which can include bandwidth, latency, and quality of service (QoS) constraints.
- **Resource Allocation:** The network allocates resources dynamically to meet user demand, ensuring adequate bandwidth, low latency, and compliance with QoS specifications.
- **Mobility Prediction:** Predicting the user's future location and mobility pattern is critical for proactive mobility management. A mLSTM model can be used to predict future user locations based on past mobility data. For the future user location, the mLSTM model predicts the following:

$$L_{t+1} = \text{LSTM}(L_t, H_t) \quad (2)$$

$$V_{t+1} = (x_{t+1} - x_t, y_{t+1} - y_t, z_{t+1} - z_t) \quad (3)$$

Where L_{t+1} represents the predicted future location, LSTM represents the mLSTM model, H_t represents the hidden state of the model at time t , and V_{t+1} represents the predicted future velocity. By incorporating existing techniques into the 5G network model, it is possible to effectively manage user mobility, make proactive handover decisions, and optimise resource allocation, thereby improving 5G environment performance and user experience.

3.2. Mobility management modelling

Modelling mobility management (MMM) for dense users in a 5G environment necessitates managing movement and handover decisions concurrently. In terms of resource allocation and handover decisions, the algorithm elaborates on the most important aspects of mobility management for these dense users.

The MMM starts by creating a dictionary called u_{md} , which will store each user's updated mobility data. For each user in the userslist, the algorithm performs the following steps:

- 1) It gets the user's mobility data (location and velocity) from the mobilitydata dictionary.
- 2) It computes the user's current distance from the serving base station using the location data and the t_d . This step decides whether a handover decision should be made.
- 3) It checks to see if the current distance exceeds the threshold distance. If the condition is met, it means the user has travelled beyond the threshold distance, necessitating a handover.

The proactive handover algorithm is activated by the algorithm to determine the user's target base station. The PH_t , as well as signal strength and available resources, are used to make the decision. The algorithm modifies the user serving base station and other handover-related parameters in the mobility data because of the handover decision. Following the handover decision, the algorithm updates the user mobility data to take their movement model into account. This entails updating the user's location and velocity based on their mobility model. The algorithm uses the RAA to allocate resources to the user. This step assesses the network's available resources and user demand requirements. Depending on the network policies, the specific resource allocation algorithm may include considerations for proportional fairness or max-min fairness. The user's u_{md} data dictionary stores updated mobility data, such as the user's updated location, velocity, serving base station, and allocated resources. The algorithm then returns the dictionary u_{md} data, which contains each user's updated mobility data. The algorithm provides a framework for modelling mobility management in a 5G environment. Variables such as user distance from the serving base station, handover decisions, mobility updates, and resource allocation are all considered. The handover algorithm, mobility model equations, and resource allocation algorithm can all be tailored to meet the needs of the network and improve the overall user experience. The position and velocity of each user i in the system are represented by $L_i = (x_i, y_i, z_i)$ at time t and $V_i = (v_{x,i,t}, v_{y,i,t}, v_{z,i,t})$. The position and velocity of a user can be updated over time using the equations below.

$$x_i(t+1) = x_i(t) + v_{x,i,t} \quad (4)$$

$$y_i(t+1) = y_i(t) + v_{y,i,t} \quad (5)$$

$$z_i(t+1) = z_i(t) + v_{z,i,t} \quad (6)$$

Each user i 's handover decision is determined by the user's distance from the serving BS and a predetermined threshold distance d_t .

$$d_i(t) = \sqrt{(x_i(t) - x_{BS,i})^2 + (y_i(t) - y_{BS,i})^2 + (z_i(t) - z_{BS,i})^2} \quad (7)$$

If $dt(i)$ is greater than d_t , a handover decision is made, and the user i must be transferred to a target BS. Proactive handover decisions aim to anticipate and initiate handovers for mobile users in advance to ensure continuous connectivity and avoid potential service disruptions. These options are determined by user predicted future locations and the QoS threshold. Let's talk about proactive handover decisions and equations for determining when to start a handover. Proactive Handover Threshold: A proactive handover threshold is defined to determine when a user's predicted future location is close to the serving BS's coverage edge. This threshold, denoted as d_p , indicates the maximum distance between the user's predicted location and the serving BS's coverage boundary.

After making a proactive handover decision, the network must choose an appropriate BS target for the user. A variety of factors, including signal strength, available resources, and proximity to nearby BSs, can all influence this decision.

The handover process is then carried out to transfer the user connection from the serving BS to the target BS. This necessitates changing the user connection parameters, releasing resources at the serving BS, and connecting to the destination BS.

The decision process demonstrates how to make proactive handover decisions based on predicted future user locations and the proactive handover threshold. The network can ensure seamless connectivity and reduce the risk of service disruptions by initiating handovers in advance when users approach the edge of the serving BS's coverage area.

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1. Normalize: `historicaldata = normalize(historicaldata)`
 2. Split: `train_set, val_set = split_data(historicaldata)`
 3. Initialize Model:
 - `model = initialize_mLSTM(numlayers, hiddenunits, dropoutrate)`
 - `compile(model, optimizer='Adam', loss='mean_squared_error')`
 4. Train Model:
 - `train(model, train_set, epochs=numepochs, learning_rate=learningrate, validation_data=val_set)`
 5. Future Mobility Prediction:
 - `predicteddata = predict_future_mobility(model, historicaldata, futuresteps)`
 6. Return: `predicteddata`
-

The network continuously predicts future locations and velocities for dense users using the mLSTM model. When a user's predicted location is near the serving BS's coverage edge, the network makes a proactive handover decision to switch the user to a target BS with better coverage. The mobility management system continuously updates user locations, makes proactive handover decisions, and allocates resources to meet user demands in a dynamic 5G environment with dense user densities, ensuring seamless connectivity and optimal network performance for all users.

3.3. MLSTM model architecture

To improve user mobility, a mLSTM model architecture is proposed for 5G networks. To capture temporal dependencies in user mobility data, this architecture makes use of deep learning and LSTM networks. Furthermore, the addition of residual connections improves the model's ability to capture complex temporal dynamics and long-term dependencies, resulting in more accurate predictions and mobility management effectiveness. The mLSTM model's architecture is made up of several key components:

Input Layer: The input layer receives historical user mobility data such as previous locations, timestamps, and other relevant attributes. This data serves as the foundation for learning and predicting users' future locations.

LSTM Layers: LSTM layers are used to capture the temporal dependencies in user mobility data. They examine the data's sequential nature, allowing the model to discover patterns and make predictions.

The residual connections within the LSTM layers enhance the model's ability to capture long-term dependencies by allowing the model to incorporate information from previous time steps more effectively.

Dense Layers: Following the LSTM layers, one or more dense layers can be added to the architecture. These layers help with the extraction of higher-level features as well as the subsequent processing of the LSTM layer learned representations. They help the model capture complex patterns and make predictions more accurately.

Output Layer: The output layer predicts future user locations based on the input data and the learned representations. The output could be probability distributions over different cells or coordinates representing the predicted locations.

The mLSTM model prediction can differ depending on the task and output representation. A common method for predicting users' future locations is to use a fully connected layer with the appropriate activation functions.

$$y_t = \text{softmax}(W_y \cdot h_t + b_y) \quad (8)$$

where W_y denotes the weights of the fully connected layer, b_y denotes the biases of the fully connected layer, and softmax denotes the probability distribution over various cells based on the predicted locations. The mLSTM model is trained using historical mobility data to predict future user locations based on the inputs provided. The training process involves minimising prediction errors by optimising model parameters such as weights and biases using techniques such as backpropagation and gradient descent. The proposed method captures both short-term and long-term dependencies in user mobility patterns, allowing for accurate predictions of future locations using this mLSTM model architecture. This enables proactive resource allocation and handover decisions in 5G networks, ultimately improving mobility management effectiveness and the overall user experience.

3.4. Deep residual mechanism for LSTM

The incorporation of residual connections within the LSTM architecture was inspired by the concept of residual learning from residual networks (ResNet) in computer vision tasks. Residual connections allow for the efficient propagation of information and gradients across multiple layers, making it easier to model complex temporal dynamics and capture long-term dependencies in user mobility data. In the LSTM architecture, input flows through a series of LSTM layers, each of which processes the input at a specific time step. The output of one LSTM layer becomes the input for the next layer, and so on. While this sequential flow of information is effective for capturing short-term dependencies, modelling complex temporal dynamics and capturing long-term dependencies may be challenging. To overcome this limitation, deep residual layers introduce skip connections, also known as residual connections, which allow the direct flow of information from one LSTM layer to a later layer, bypassing intermediate layers. These residual connections provide an additional path for information propagation, allowing the model to capture long-term dependencies more effectively. The idea behind residual connections is to find the difference between a layer's input and output. The residual connection skips the intermediate layers' computation and adds the original input directly to the layer output. This helps to alleviate the problem of vanishing or exploding gradients in deep neural networks. By incorporating residual connections in LSTM layers, the mLSTM model can effectively capture complex temporal dynamics. The residual connections allow the model to retain information from previous time steps, enhancing its ability to capture long-term dependencies and overcoming the challenges posed by user mobility patterns characterised by irregular movements, sudden changes, or random cell transitions.

3.5. Deep residual modelling

Skip connections that bypass intermediate LSTM layers are included in mLSTM connections. Consider the following simplified scenario with two LSTM layers: the expression for the mLSTM connections is,

$$h_1 = \text{LSTM}_1(x_t, h_{t-1}) \quad (9)$$

$$r = h_1 + x_t \quad (10)$$

$$h_2 = \text{LSTM}_2(r, h_{t-1}) \quad (11)$$

where x_t is the input at time t , h_{t-1} is the previous hidden state, h_1 is the first LSTM output, and r is the residual connection $= h_1 + x_t$, h_2 - second LSTM output. This facilitates the capture of long-term dependencies and aids in the preservation of critical information from earlier time steps. The concept of deep residual connections can be expanded in deeper architectures with more LSTM layers by introducing additional residual connections between layers. Deep residual connections in LSTM architectures effectively propagate information and gradients through multiple layers, allowing the model to capture complex temporal dynamics while also improving its ability to capture long-term dependencies. These connections reduce performance degradation in very deep LSTM architectures and improve the model's ability to predict future user locations and mobility patterns.

4. Results and discussion

This study creates a simulation scenario in MATLAB with the parameters listed in Table 2 to evaluate the performance of the proposed mLSTM model for improving mobility management in 5G networks.

10 BSs were chosen to represent a network with sufficient coverage and capacity for a 1 sq. km area, mimicking a dense urban or suburban setting. The chosen area of 1 sq. km provides a realistic scale for evaluating the performance of the mobility management system in a moderately-sized geographical region. With 500 users, the simulation captures a significant number of mobile devices, creating a scenario that reflects a busy network with diverse mobility patterns. The 1-hour simulation duration allows for a comprehensive analysis of the mLSTM model's effectiveness over an extended period, considering varying mobility patterns and network conditions. The mLSTM model is configured with 3 LSTM layers and 64 hidden units. These values strike a balance between model complexity and computational efficiency, aiming for optimal performance in capturing mobility patterns. A dropout rate of 0.2 is applied to prevent overfitting and enhance the generalization ability of the mLSTM model during training. A learning rate of 0.001 is chosen to control the step size during optimization. Training over 50 epochs strikes a balance between model convergence and computational efficiency. The Random Waypoint mobility model with a

minimum speed of 1 m/s, maximum speed of 5 m/s, and a pause time range of 0–30 seconds introduces realistic user mobility patterns, reflecting the dynamic nature of users in real-world scenarios.

The comparison of Proposed mLSTM with SbDMM, RL, Adaptive TTT, InterSlice, NPAHA, MBMMS, MIMM, and ACSA is based on the evaluation of various mobility management techniques.

- **InterSlice Mobility Management (ISMM)** [23]: It allows mobile devices to stay connected while switching between different slices or data plans. It focuses on slice migration during mobility events, ensuring uninterrupted connections to remote applications and services.
- **Reinforcement Learning (RL)** [24]: It is an artificial intelligence technique used for solving complex optimization problems. In the context of handover management in ultra-dense small cell networks, RL algorithms aim to optimize decision-making in dynamically complex environments.
- **Adaptive Time-to-Trigger (Adaptive TTT)** [22]: Adaptive TTT is introduced in intelligent handover procedures, adding adaptive time-to-trigger and hysteresis margin to traditional algorithms.
- **Stackelberg Game Approach (SbDMM)** [26]: It is applied to multiple resource allocation and pricing in mobile edge computing. It involves a decision-making strategy where a leader (service provider) determines resource allocation prices, influencing followers (users) seeking to maximize their gains.
- **Neighbor predictive adaptive handoff algorithm (NPAHA)** [5]: It is a technique used in cellular networks to improve handoff decisions for mobile devices. It uses information from neighboring base stations to anticipate handoff needs and adjusts handoff parameters accordingly, leading to more efficient and seamless transitions between base stations for mobile devices.
- **Multihoming-based mobility management scheme (MBMMS)** [8]: It is a method of managing mobile devices as they move between different networks. It utilizes multiple interfaces on a device to establish simultaneous connections to multiple networks, allowing for seamless handovers and improved network performance. This approach reduces latency and improves network availability for mobile users.
- **Media independent mobility management (MIMM)** [10]: It is a framework that enables seamless roaming and handover of mobile devices across different network technologies, such as Wi-Fi, cellular, and satellite. It ensures uninterrupted connectivity and transparent movement of devices without the need for manual network reconfiguration, improving the user experience.
- **Adaptive cell selection algorithm (ACSA)** [20]: It is a method for selecting the most suitable communication channel in a cellular network. It uses parameters such as signal strength, traffic load, and error rate to intelligently choose the best cell for a given device, resulting in improved network performance and efficiency.

Table 2
Simulation Parameters.

Network Setup	Values
Number of BSs	10
Area covered	1 sq. km
Number of users	500
Simulation duration	1 hour
mLSTM Model	
Number of LSTM layers	3
LSTM hidden units	64
Dropout rate	0.2
Learning rate	0.001
Number of training epochs	50
User Mobility Model	
Mobility Model	Random Waypoint
Minimum speed	1 m/s
Maximum speed	5 m/s
Pause time range	0–30 seconds

4.1. Estimation of packet loss rate

The Packet Loss rate for mobility management in 5G networks refers to the percentage of data packets that are unsuccessfully delivered or received during the handover process, where a mobile device switches between different base stations while maintaining the connection to the network. In 5G networks, the goal is to minimize packet loss during mobility management to maintain seamless connectivity for mobile devices. This is achieved through advanced techniques such as fast handovers, improved signaling procedures, and intelligent traffic routing. Low packet loss rates are crucial in providing a smooth and uninterrupted user experience in 5G networks, especially in scenarios where high mobility is expected. Table 3 shows the estimation of packet loss rate.

Fig. 2 shows the estimation of packet loss rate. In a computation

Table 3
Packet Loss Rate (in %).

Users	InterSlice	Adaptive TTT	RL	SbDMM	NPAHA	MBMMS	MIMM	ACSA	Proposed mLSTM
50	1	0.8	0.6	0.4	2.6	2.8	2.2	2.2	0.2
100	1.3	1	0.7	0.6	2.3	2.6	2.1	2	0.3
150	1.4	1.1	0.8	0.7	2.2	2.5	2	1.9	0.4
200	1.5	1.2	0.9	0.8	2.1	2.4	1.9	1.8	0.5
250	1.5	1.2	0.9	0.8	2.1	2.4	1.9	1.8	0.5
300	1.6	1.3	1	0.9	2	2.3	1.8	1.7	0.6
350	1.7	1.4	1.1	1	1.9	2.2	1.7	1.6	0.7
400	1.8	1.5	1.2	1	1.8	2.1	1.6	1.6	0.7
450	1.9	1.6	1.3	1.1	1.7	2	1.5	1.5	0.8
500	2.2	1.8	1.4	1.3	1.4	1.8	1.4	1.3	0.9

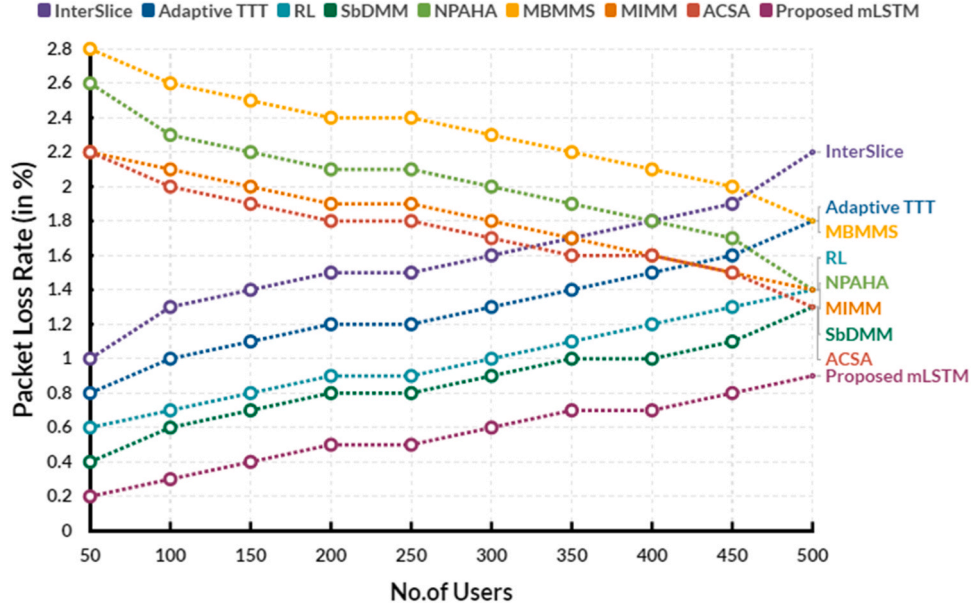


Fig. 2. Packet Loss Rate (in %).

cycle, InterSlice obtained 2.22 %, Adaptive TTT reached 1.8 %, RL obtained 1.4 %, SbDMM reached 1.3 %, NPAHA obtained 1.4 %, MbMMS reached 1.8 %, MIMM reached 1.4 %, ACSA obtained 1.3 % packet loss rate. The proposed mLSTM obtained 0.9 % packet loss rate.

4.2. Estimation of handover success rate

The Handover Success Rate is a metric used to evaluate the effectiveness of mobility management in 5G networks. It measures the percentage of handovers successfully completed without interruptions or delays during a mobile device's movement between different network cells. In a 5 G network, efficient mobility management is crucial due to the high data speeds and low latency requirements. This involves seamless handovers between cells as the device moves, ensuring

uninterrupted service and an optimal user experience. A high HOSR indicates that the network's mobility management works effectively, providing reliable connectivity for mobile devices in 5G networks. Table 4 shows the estimation of HSR.s

Fig. 3 shows the estimation of Handover Success Rate. In a computation point, InterSlice obtained 87.8 %, Adaptive TTT reached 89.2 %, RL obtained 91.5 %, SbDMM reached 92.3 %, NPAHA obtained 75.5 %, MbMMS reached 74.00 %, MIMM reached 61.3 %, ACSA obtained 70.3 % Handover Success Rate. The proposed mLSTM obtained 95.1 % Handover Success Rate.

4.3. Estimation of handover latency

Handover latency refers to the time it takes for a mobile device to

Table 4
Handover Success Rate (HSR).

Users	InterSlice	Adaptive TTT	RL	SbDMM	NPAHA	MBMMS	MIMM	ACSA	Proposed mLSTM
50	80.2	82.6	84.5	85.8	83.1	80.6	68.3	76.8	88.9
100	80.9	83.1	85.2	86.4	82.4	80.1	67.6	76.2	89.5
150	81.6	83.9	85.9	87.2	81.7	79.3	66.9	75.4	90.7
200	82.1	84.3	86.7	87.6	81.2	78.9	66.1	75.0	91.2
250	82.9	85.3	87.1	88.6	80.4	77.9	65.7	74.0	92.1
300	83.2	85.6	87.4	88.7	80.1	77.6	65.4	73.9	92.3
350	85.1	87.6	88.9	90.2	78.2	75.6	63.9	72.4	93.4
400	85.3	87.9	89.1	90.8	78.0	75.3	63.7	71.8	93.7
450	86.1	88.3	90.2	91.5	77.2	74.9	62.6	71.1	94.6
500	87.8	89.2	91.5	92.3	75.5	74.0	61.3	70.3	95.1

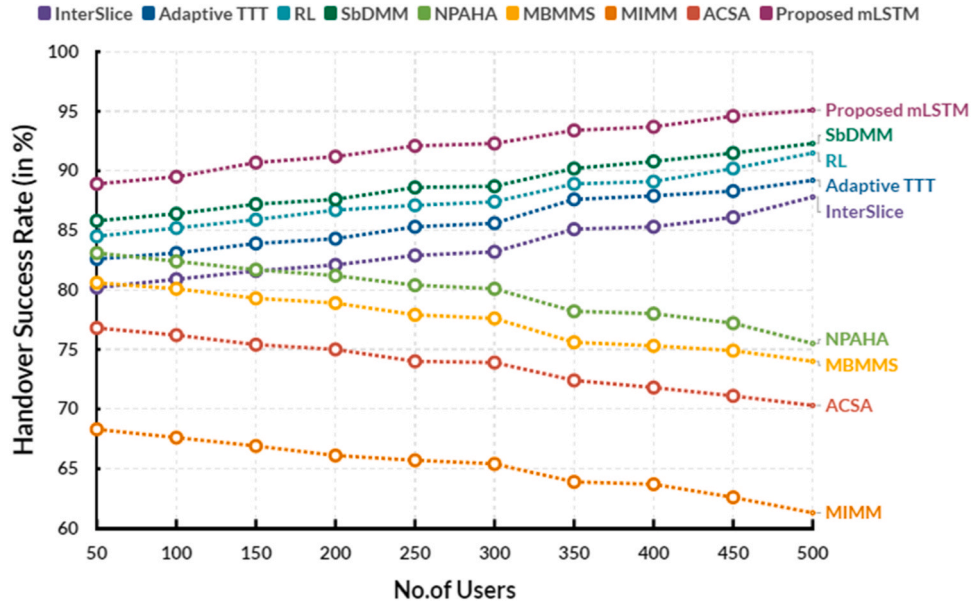


Fig. 3. Handover Success Rate (HSR).

switch between different base stations or cells while maintaining a continuous connection to the network. In 5G networks, the handover process has been optimized to reduce latency to fewer than ten milliseconds, ensuring seamless and uninterrupted user services. This is achieved through improved handover signaling protocols and advanced technologies such as Beamforming and multi-connectivity. 5G networks also use predictive handover algorithms based on real-time network and device information to initiate handovers, proactively reducing latency. Overall, the reduced handover latency in 5G networks allows for faster and more efficient mobility management, enabling better user experiences for mobile services. Table 5 shows the handover latency.

Fig. 4 shows the handover latency. In a computation point, the InterSlice obtained 71 ms, Adaptive TTT reached 68 ms, RL obtained 65 ms, SbDMM reached 63 ms, NPAHA obtained 61 ms, MBMMS reached 63 ms, MIMM reached 64 ms, ACSA obtained 62 ms handover latency. The proposed mLSTM obtained 57 ms handover latency.

4.4. Estimation of mobility prediction accuracy

Mobility prediction accuracy in 5G networks refers to the network's ability to predict a mobile device's future location. This is important for effective mobility management, which involves seamless handoffs between different base stations and optimal routing of network resources. In order to achieve high accuracy, 5G networks use advanced techniques such as machine learning, data analytics, and real-time tracking of user behaviors. These techniques consider factors such as user mobility patterns, network load, and environmental conditions to predict a user's location accurately. This results in faster and more efficient handoffs and

resource allocation, thereby improving the overall performance and user experience in 5G networks. Table 6 shows the estimation of mobility prediction accuracy.

Fig. 5 shows the mobility prediction accuracy. In a computation point, the InterSlice obtained 82.9 %, Adaptive TTT reached 84.7 %, RL obtained 86.1 %, SbDMM reached 87.3 %, NPAHA obtained 72.7 %, MBMMS reached 79.5 %, MIMM reached 84.9 %, ACSA obtained 86.1 % Mobility Prediction Accuracy. The proposed mLSTM obtained 91.2 % Mobility Prediction Accuracy.

4.5. Estimation of throughput

Throughput for mobility management in 5G networks refers to the speed at which data can be transferred between devices during handovers between different network cells. Handovers occur when a user moves from one cell to another, and the device needs to connect to a new base station. In 5G networks, the throughput for mobility management is enhanced through Beamforming, which directs signals toward the user's device, and multi-connectivity, which allows for simultaneous connections to multiple base stations. This improves the reliability and efficiency of handovers, reducing latency and ensuring uninterrupted connectivity, which is crucial for applications like autonomous vehicles and virtual reality. Overall, the higher throughput for mobility management in 5G networks enables faster and smoother interaction between devices and the network. Table 7 shows the estimation of throughput.

Fig. 6 shows the estimation of throughput. In a computation point, the InterSlice obtained 49.5 MBps, Adaptive TTT reached 51.3 MBps, RL

Table 5
Handover Latency (in ms).

Users	InterSlice	Adaptive TTT	RL	SbDMM	NPAHA	MBMMS	MIMM	ACSA	Proposed mLSTM
50	50	47	44	42	40	42	43	41	36
100	52	49	46	42	42	44	45	41	38
150	54	51	48	46	44	46	47	45	40
200	56	53	50	48	46	48	49	47	42
250	58	55	52	50	48	50	51	49	44
300	60	56	54	52	50	51	53	51	45
350	62	59	56	54	52	54	55	53	48
400	64	61	58	56	54	56	57	55	50
450	66	63	60	58	56	58	59	57	52
500	71	68	65	63	61	63	64	62	57

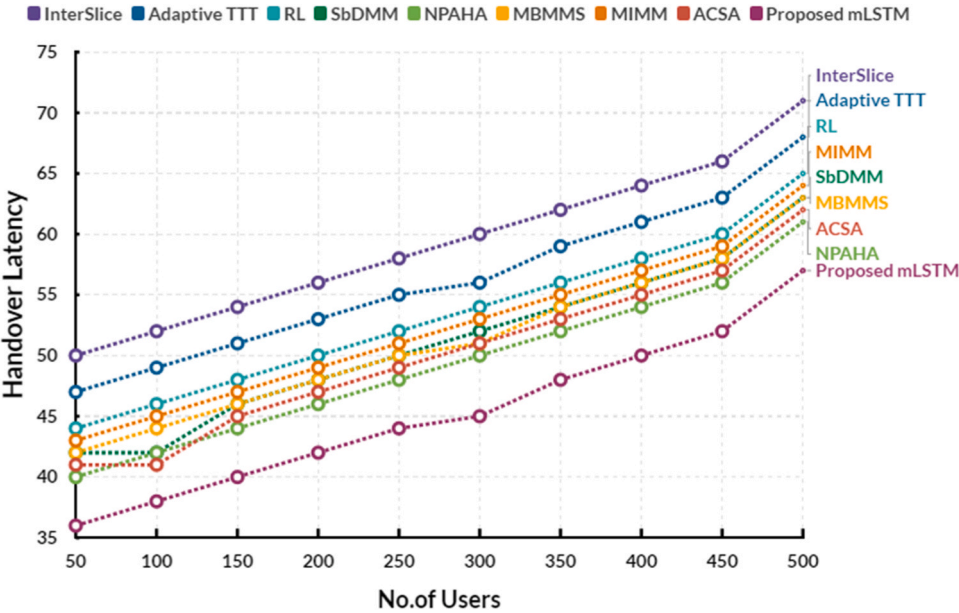


Fig. 4. Handover Latency (in ms).

Table 6
Mobility Prediction Accuracy (in %).

Users	InterSlice	Adaptive TTT	RL	SbDMM	NPAHA	MBMMS	MIMM	ACSA	Proposed mLSTM
50	70.6	72.8	74.2	75.1	60.4	67.6	73.0	73.9	79.3
100	73.1	75.3	76.3	77.6	62.9	70.1	75.1	76.4	81.7
150	73.9	76.2	77.1	78.4	63.7	71.0	75.9	77.2	82.5
200	74.6	76.6	77.9	79.2	64.4	71.4	76.7	78.0	83.9
250	76.7	78.6	79.9	81.2	66.5	73.4	78.7	80.0	85.1
300	77.9	79.8	81.1	82.4	67.7	74.6	79.9	81.2	86.3
350	78.5	80.3	81.8	83.1	68.3	75.1	80.6	81.9	87.6
400	79.6	81.4	82.9	84.2	69.4	76.2	81.7	83.0	88.7
450	81	82.9	84.2	85.5	70.8	77.7	83.0	84.3	89.4
500	82.9	84.7	86.1	87.3	72.7	79.5	84.9	86.1	91.2

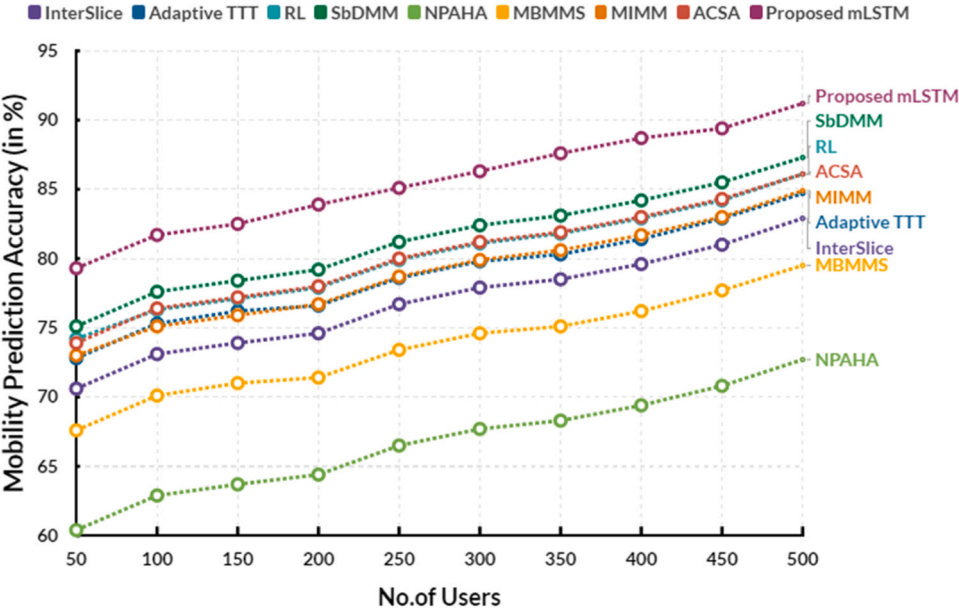


Fig. 5. Mobility Prediction Accuracy (in %).

Table 7
Throughput (in Mbps).

Users	InterSlice	Adaptive TTT	RL	SbDMM	NPAHA	MBMMS	MIMM	ACSA	Proposed mLSTM
50	31.8	33.1	34	34.6	13.8	15.1	16.0	16.6	36.9
100	33.5	34.8	35.9	36.2	15.5	16.8	17.9	18.2	38.6
150	34.9	36.2	37.3	38	16.9	18.2	19.3	20.0	40.3
200	37.1	38.4	39.6	40.2	19.1	20.4	21.6	22.2	42.8
250	38.4	39.9	40.7	41.9	20.4	21.9	22.7	23.9	44.6
300	39.2	40.5	41.6	42.8	21.2	22.5	23.6	24.8	45.2
350	42.2	43.8	44.6	46	24.2	25.8	26.6	28.0	48.9
400	43.4	44.9	45.8	47.2	25.4	26.9	27.8	29.2	50.1
450	45.4	47	48	49.4	27.4	29.0	30.0	31.4	52.5
500	49.5	51.3	52.6	53.8	31.5	33.3	34.6	35.8	57.3

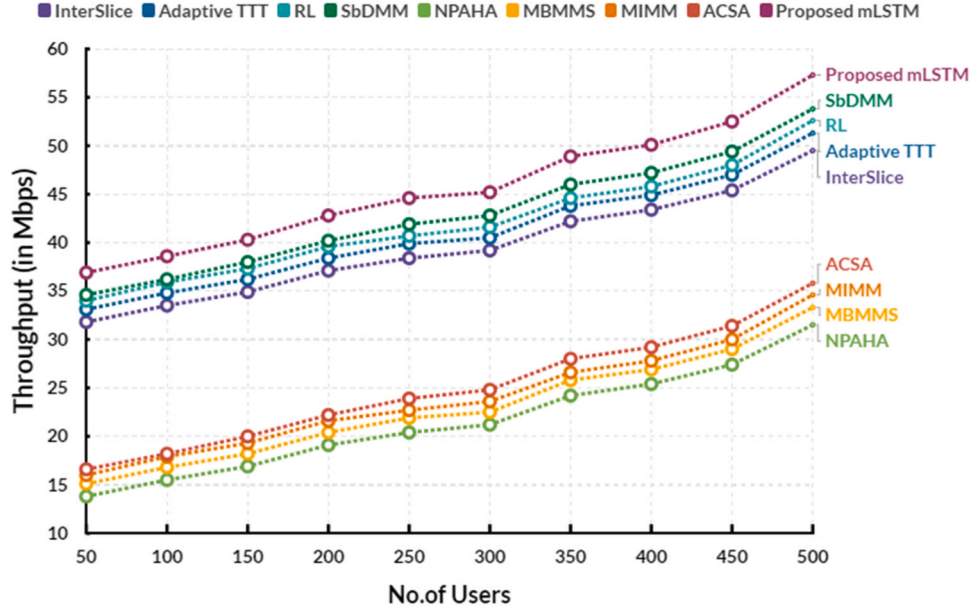


Fig. 6. Throughput (in Mbps).

obtained 52.6 MBps, SbDMM reached 53.8 MBps, NPAHA obtained 31.5 MBps, MBMMS reached 33.3 MBps, MIMM reached 34.6 MBps, ACSA obtained 35.8 MBpsThroughput. The proposed mLSTM obtained 57.3 MBpsThroughput.

4.6. Estimation of latency

Latency for mobility management in 5G networks refers to the delay when a 5G device moves from one location to another and needs to connect with a new base station. This process involves multiple steps, including identifying the new base station, authentication, and setting up a new session. With 5G technology, latency is significantly reduced compared to previous generations, as low as one millisecond. This

allows for seamless and uninterrupted connectivity, which is essential for applications. Table 8 shows the estimation of latency.

Fig. 7 shows the estimation of latency. In a computation point, the InterSlice obtained 16.3 ms, Adaptive TTTreached 17.5 ms, RL obtained 18.3 ms, SbDMM reached 19.6 ms, NPAHA obtained 8.3 ms, MBMMS reached 9.5 ms, MIMM reached 10.3 ms, ACSA obtained 11.6 ms Latency. The proposed mLSTM obtained 20.9 msLatency.

5. Ablation study

The number of LSTM layers and hidden units is crucial in enhancing mobility management in 5G networks using the deep residual LSTM model. These layers act as memory units for the network to retain

Table 8
Latency (in ms).

Users	InterSlice	Adaptive TTT	RL	SbDMM	NPAHA	MBMMS	MIMM	ACSA	Proposed mLSTM
50	10.7	11.4	12	13.2	2.7	3.4	4.0	5.2	14.6
100	11.9	12.6	13.3	14.6	3.9	4.6	5.3	6.6	15.9
150	12.4	13.2	13.9	15.1	4.4	5.2	5.9	7.1	16.5
200	12.6	13.4	14.1	15.3	4.6	5.4	6.1	7.3	16.7
250	13.5	14.3	15	16.2	5.5	6.3	7.0	8.2	17.6
300	14.2	15.4	16.1	17.2	6.2	7.4	8.1	9.2	18.7
350	14.8	15.4	16.2	17.5	6.8	7.4	8.2	9.5	18.8
400	15.2	16.3	17	18.2	7.2	8.3	9.0	10.2	19.6
450	15.9	16.4	17.2	18.5	7.9	8.4	9.2	10.5	19.8
500	16.3	17.5	18.3	19.6	8.3	9.5	10.3	11.6	20.9

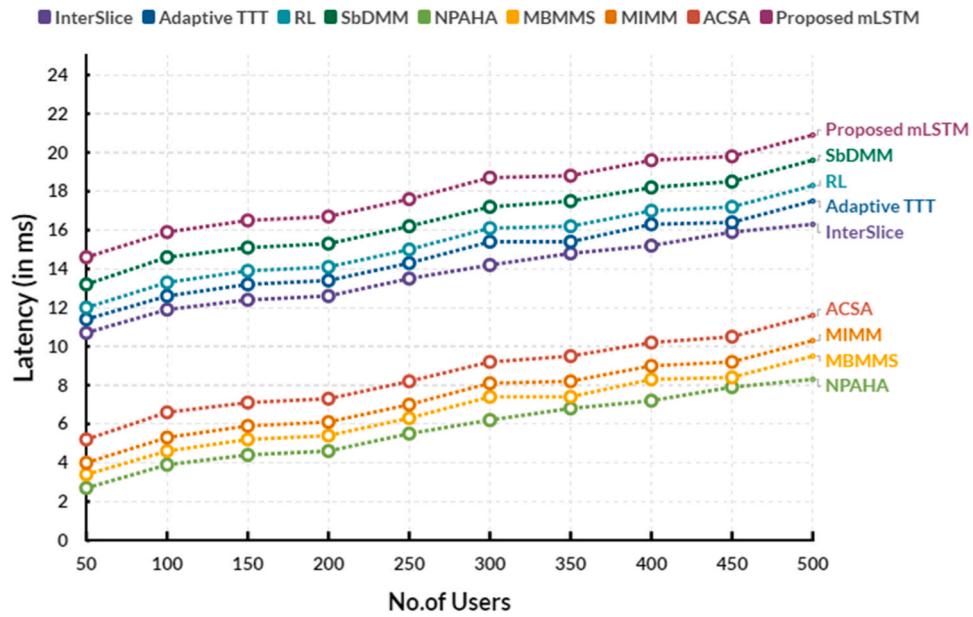


Fig. 7. Latency (in ms).

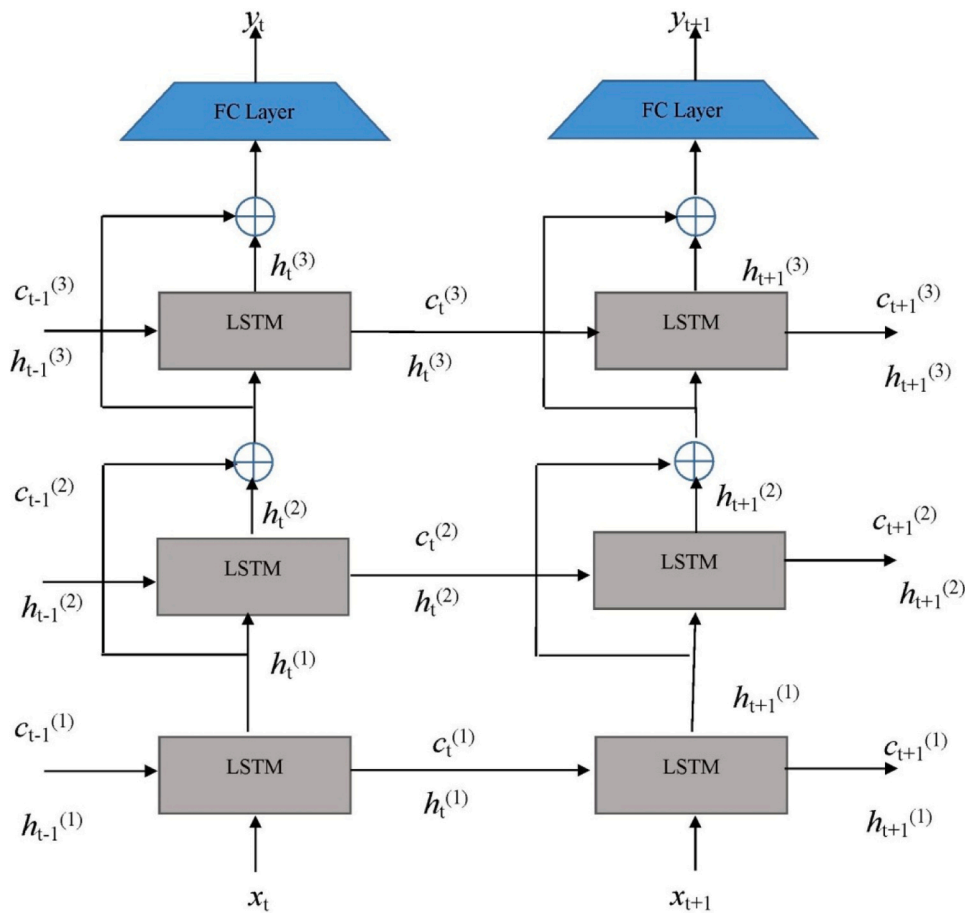


Fig. 8. (a). Performance of proposed mLSTM model (3-Layer Network). (b). Performance of proposed mLSTM model (2-Layer Network).

information over time, while the hidden units perform complex computations on the input data. Fig. 8(a) shows the performance of proposed mLSTM model (3-Layer Network) and Fig. 8(b) shows the performance

of proposed mLSTM model (2-Layer Network).

The proposed mLSTM model consists of four layers: three LSTM layers and one fully connected layer. The first LSTM layer takes in three

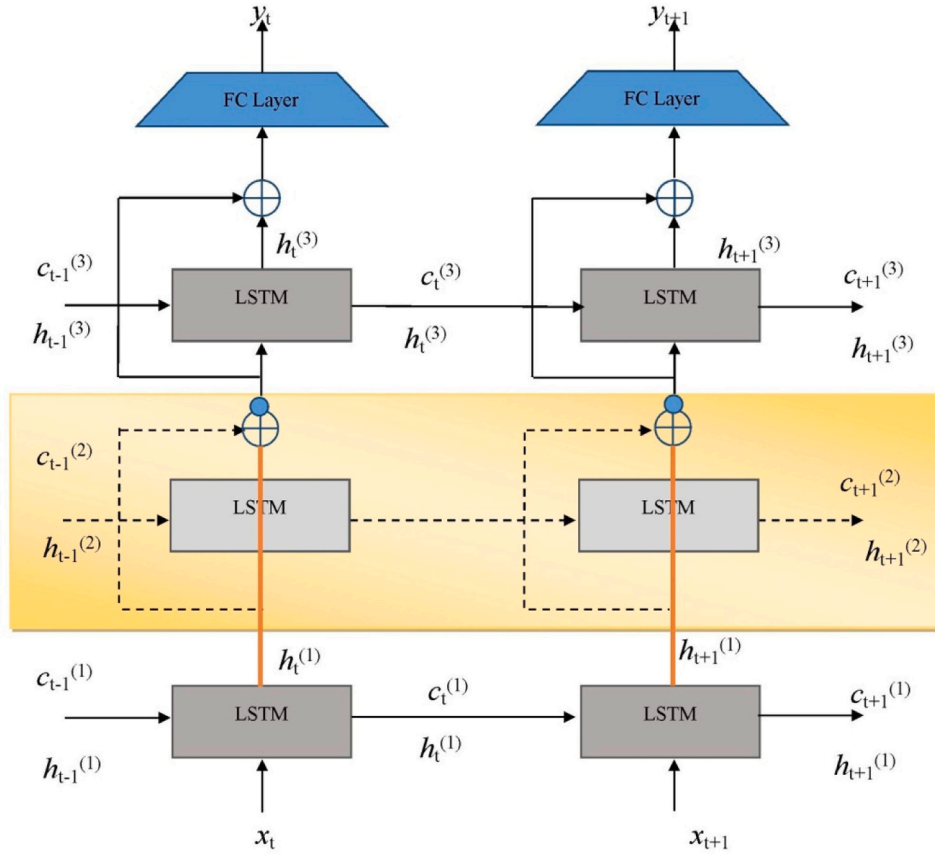


Fig. 8. (continued).

inputs: the hidden state $h_{t-1}^{(1)}$ and cell state $c_{t-1}^{(1)}$ from the previous time step, as well as the current input x_t . The LSTM layer then processes these inputs and produces two outputs: the new hidden state $h_t^{(1)}$ and cell state $c_t^{(1)}$ at the current time step. This process is repeated for each time step in the input sequence, allowing the model to capture long-term dependencies in the data. The second LSTM layer follows a similar structure, taking in the previous hidden state $h_{t-1}^{(2)}$ and cell state $c_{t-1}^{(2)}$ from the first layer, as well as the hidden state $h_t^{(1)}$ from the previous layer. This allows the model to incorporate information from both the previous time step and the previous layer. Again, the LSTM layer produces two outputs: the new hidden state $h_t^{(2)}$ and cell state $c_t^{(2)}$ at the current time step. The third layer operates in the same manner, taking in the previous hidden state $h_{t-1}^{(3)}$ and cell state $c_{t-1}^{(3)}$ from the second layer, as well as the hidden state $h_t^{(2)}$ from the previous layer. This allows the model to capture even higher-level dependencies by incorporating information from multiple previous layers. The LSTM layer produces two outputs: the new hidden state $h_t^{(3)}$ and cell state $c_t^{(3)}$ at the current time step. The output of the third LSTM layer is then passed to a fully connected layer, which produces the final predicted output y_t . This layer uses weights that are learned during training to map the LSTM outputs to the desired output format. The proposed mLSTM model also includes a separate fully connected layer for predicting timesteps $t+1$. This layer takes in the hidden state $h_{t+1}^{(3)}$ from the third layer and produces the predicted output for the next time step, allowing the model to make sequential predictions. By incorporating multiple LSTM layers, the proposed mLSTM model is able to capture both short-term and long-term dependencies in the data. This is achieved by utilizing information from multiple previous time steps and previous layers, allowing the model to learn complex patterns in the data. The addition of the fully connected layers also adds flexibility to the model, allowing for the prediction of both the current and future timesteps. Decreasing the number of layers and hidden units may result in the model needing more capacity to capture

complex patterns and may decrease its effectiveness. In Fig. 8(b) the second LSTM layer has been removed. The effects of ablation have been shown in the following Table 9.

Packet Loss Rate represents the percentage of data packets lost during transmission. In the proposed mLSTM model in 3-Layer Network, the PLR was 0.9 %, meaning that only 0.9 % of the data packets were lost during transmission. In, 2-Layer Network, the PLR increased to 1.22 %, indicating a slight increase in data packet loss.

The Handover Success Rate refers to the measure of successful handovers between different parameters. The proposed mLSTM model had a success rate of 95.1 % (3-Layer Network), which decreased to 86.24 % in 2-Layer Network. This indicates that the removed LSTM layer played a significant role in the success of parameter handovers.

Handover Latency refers to the time delay in transferring information between different memory cells. The proposed mLSTM model has a handover latency of 57 ms in 3-Layer Network, which is the time needed to transfer parameters between memory cells. In 2-Layer Network, handover latency increases to 65.54 ms due to the removal of the second LSTM layer in the model. This increase in latency may affect the overall

Table 9
Ablation based Performance Comparison.

Parameters	Proposed mLSTM (3-Layer Network)	Proposed mLSTM (2-Layer Network)
Packet Loss Rate (in %)	0.9	1.22
Handover Success Rate (in %)	95.1	86.24
Handover Latency (in ms)	57	65.54
Mobility Prediction Accuracy (in %)	91.2	80.62
Throughput (in Mbps)	57.3	53.03
Latency (in ms)	20.9	12.79

performance of the model.

The Mobility Prediction Accuracy is a measure of how well an mLSTM algorithm can predict movement trajectories. By removing the second layer, the initial proposed mLSTM achieved an accuracy of 91.2 % in 3-Layer Network. However, after removing the second layer, the accuracy decreased to 80.62 % in 2-Layer Network. This suggests that the second LSTM layer was essential for improving the prediction accuracy.

Throughput refers to the speed at which data can be processed and transmitted within a system. In this context, the proposed mLSTM in 3-Layer Network achieved a speed of 57.3 Mbps, while the proposed mLSTM in 2-Layer Network achieved a slightly lower speed of 53.03 Mbps. This indicates that the removal of LSTM layers may have slightly reduced the efficiency of the system in processing and transmitting data.

Latency refers to the delay in the processing of an operation. In the proposed mLSTM method, the initial version had a latency of 20.9 ms in 3-Layer Network, but after performing ablation, it was reduced to 12.79 ms in 2-Layer Network. This improvement in latency means that the proposed mLSTM method can process operations faster and more efficiently, making it a more efficient and optimized solution.

In mLSTM networks, the residual connections allow information from previous time steps to be preserved and passed on to the current time step, improving the network's ability to capture long-term dependencies in the data. The ablation experiment for mLSTM networks would involve training and evaluating the network with and without residual connections. This would allow us to compare the network's performance with and without this feature and determine its contribution to the overall performance.

Fig. 9(a) shows the mLSTM networks without residual connections. In this, the mLSTM network could learn and make predictions. Still, it would likely need help with longer sequences and have a more challenging time capturing relationships between distant time steps. This is because, in the absence of residual connections, the gradient of the loss function decays exponentially as it propagates through the network, making it difficult for the network to learn from distant time steps.

Fig. 9(b) shows the mLSTM networks with residual connections. Here, the mLSTM network with residual connections could mitigate this issue by allowing the gradient to flow more quickly through the network. The residual connections also help to prevent vanishing

Table 10

Ablation performance (With and without Residual Layers).

Parameters	Performance of mLSTM (without Residual Connection)	Performance of mLSTM (with Residual Connection)
Packet Loss Rate (in %)	26.54	0.9
Handover Success Rate (in %)	68.79	95.1
Handover Latency (in ms)	81.33	57
Mobility Prediction Accuracy (in %)	47.61	91.2
Throughput (in Mbps)	39.88	57.3
Latency (in ms)	9.27	20.9

gradients, enabling the network to effectively learn from distant time steps and capture long-term dependencies in the data. Table 10 shows the performance of mLSTM network with and without residual connections.

Parameters in a mLSTM network without residual connection refer to the weights and biases used in training and prediction. These parameters learn the relationships and patterns between input data and corresponding outputs. The mLSTM without residual connection had a higher packet loss rate of 26.54 % compared to 0.9 % for the mLSTM with residual connection. Adding a residual connection can help reduce packet loss during transmission. The mLSTM with residual connection had a much higher success rate of 95.1 % compared to 68.79 % for the mLSTM without residual connection. This indicates that the residual connection can improve network stability and reliability. The mLSTM with residual connection had a lower handover latency of 57 ms compared to 81.33 ms for the mLSTM without residual connection. This suggests that the residual connection can help reduce handover delay and improve network efficiency. The mLSTM with residual connection had a significantly higher accuracy of 91.2 % compared to 47.61 % for the mLSTM without residual connection. This shows that the residual connection can improve the ability to predict and handle mobile devices' movements. The mLSTM with residual connection had a higher throughput of 57.3 Mbps compared to 39.88 Mbps for the mLSTM without residual

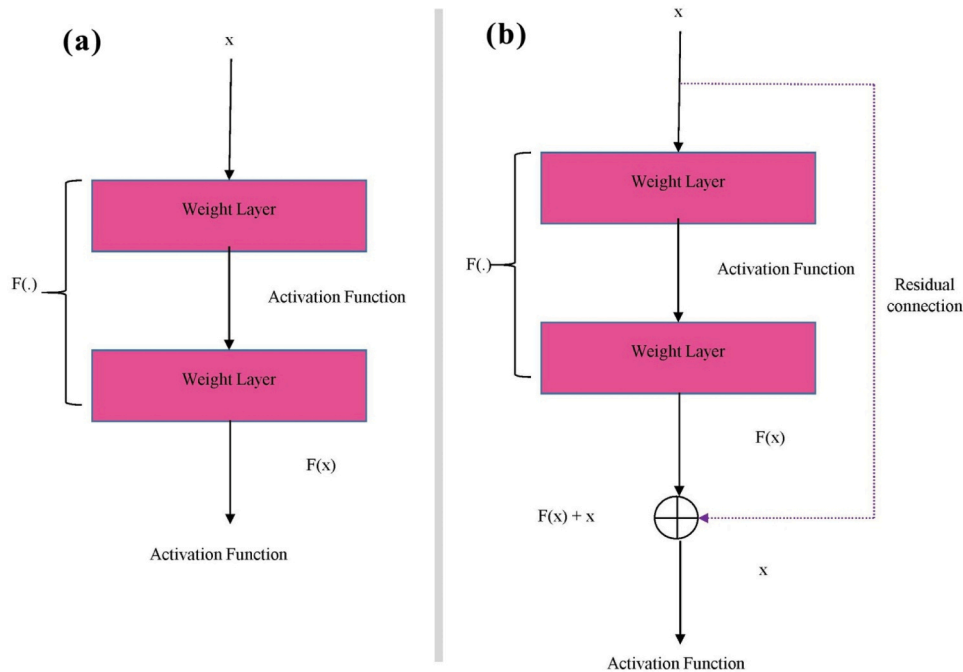


Fig. 9. (a) – mLSTM networks without residual connections; (b) – mLSTM networks with residual connections.

connection. This indicates that the residual connection can help improve the overall data transfer rate of the network. The mLSTM with residual connection had a slightly higher latency of 20.9 ms compared to 9.27 ms for the mLSTM without residual connection. This suggests that the residual connection may introduce a slight increase in latency, but overall, it does not significantly impact network performance. The ablation experiment for residual connections in LSTM networks would provide insights into the importance of this feature in capturing long-term dependencies in sequential data. It would demonstrate the network's superior performance with residual connections and emphasize its critical role in improving the overall performance of LSTM networks.

6. Conclusion

This study proposed a mLSTM model to improve mobility management in 5G networks. The results of the evaluation demonstrated that the proposed method is effective at improving various performance metrics. The proposed mLSTM obtained 0.9 % packet loss rate, 95.1 % Handover Success Rate, 57 ms handover latency, 91.2 % Mobility Prediction Accuracy, 57.3 MBps Throughput, 20.9 ms Latency. These findings demonstrate the efficacy of the proposed method for improving mobility management in 5G networks, which provides improved handover success, lower latency, higher accuracy in mobility prediction, higher throughput, and lower packet loss. Significant improvements in these metrics show that deep learning techniques, specifically the mLSTM model, have the potential to improve mobility management in 5G networks. The proposed method allows for proactive resource allocation and handover decisions based on accurate predictions of user mobility patterns, thereby optimising network performance, and improving user experience. The performance of the proposed method can be thoroughly evaluated by conducting evaluations with a larger number of users and a more extensive network configuration. Incorporating dynamic factors into the mobility management model, such as fluctuating network loads and varying channel conditions, can improve its efficacy even further.

CRediT authorship contribution statement

Abdullah Baz (Principal Investigator): Writing – review & editing, Writing – original draft, Software, Formal analysis, Conceptualization. **Jaganathan Logeshwaran**: Writing – review & editing, Writing – original draft, Software, Formal analysis, Conceptualization. **Yuvaraj Natarajan**: Writing – review & editing, Writing – original draft, Software, Formal analysis, Conceptualization. **Shobhit K. Patel**: Writing – review & editing, Writing – original draft, Software, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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